
ANEESHA AVASTHI

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EDUCATION

- PhD: Physics | JN Tata Scholar** **08/2020 - Current**
Case Western Reserve University (CWRU) | **GPA: 4.0/4.0** **Cleveland, USA**
 - Thesis: Development and Characterization of Proportional Gas Detectors for Rare Event Searches
- MSc: Physics | INSPIRE Scholar** **07/2017-05/2019**
Indian Institute of Technology Guwahati (IIT-G) | **GPA: 8.3/10.0** **Guwahati, India**
 - Thesis: Sub-wavelength Light Manipulation via Lens-Focused Phase Extraction
- BSc: Physics | INSPIRE Scholar** **07/2014 – 05/2017**
SGTB Khalsa College, University of Delhi (DU) | **Percentage: 91.36%** **Delhi, India**
 - Awarded: Best Female Student in Physics
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TEACHING EXPERIENCE

- Teaching Assistant (Departments of Physics)** **2019-2021**
State University of New York at Buffalo **Buffalo NY, USA**
Case Western Reserve University **Cleveland OH, USA**
 - Designed and delivered physics labs for 200+ undergraduates across two courses.
 - Adapted instruction to hybrid/virtual settings during COVID-19 using simulations and live demonstrations.
 - Held tutorials and office hours to mentor students, improving confidence and performance.
- Village Teaching Volunteer** **2017-2019**
Indian Institute of Technology (IIT) **Guwahati, Assam, India**
 - Taught math and science to children (ages 5–15) of campus staff with limited resources.
 - Used simple, low-cost experiments and everyday examples to spark curiosity.
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MENTORSHIP & LEADERSHIP

- Grant Reviewer**, Graduate Student Scholarship and Creative Endeavors (GSSCE) – Evaluated funding proposals for academic and creative research (2024)
 - Committee Member**, International Student Advisory Council (ISAC) – Contributed to campus-wide advocacy and support initiatives for international students (2023)
 - Mentor & Mentee**, Mather Mentorship Program for Women in STEM – Participated in peer and cross-level mentorship to support women pursuing STEM careers (2023–2025)
 - Treasurer**, Graduate Council of Arts & Sciences (GCAS) – Managed finances and budget planning for student-led events and initiatives (2022–2024)
 - First-Year Mentor**, Physics Underrepresented Minority Students (PURMS) – Provided academic and social mentorship for incoming minority students in physics (2021–2025)
 - Judge & Reviewer**, Intersections: Undergraduate Poster Symposium – Reviewed and evaluated student research posters across physics and interdisciplinary topics (2020–2025)
 - Organizer/President**, Local Inspirational Figures Talk (LIFT) – Amplified voices of students from the Sciences, Humanities and Arts.
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SCIENCE COMMUNICATION AND OUTREACH

- Guided a Cleveland high school student (first-generation) exploring pathways to college.
 - Presented at the annual 3 Minute Thesis (3MT) contest.
 - Organized public telescope sessions, guiding students through their first views of Jupiter's moons.
 - Presented a poster at the Innovation Week contest to describe my research to lay-people.
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PUBLICATIONS

- A. Anker, A. Avasthi, *et al.*, Report From The Workshop On Xenon Detector 0 $\nu\beta\beta$ Searches: Steps Towards The Kilotonne Scale, arXiv:2404.19050 **(2024)**
- A. Avasthi, T. Bezerra, *et al.*, Large Low Background kTon-Scale Liquid Argon Time Projection Chambers, arXiv:2301.11878 **(2023)**
- A. Avasthi, T.W. Bowyer, *et al.*, Kilotonne-Scale Xenon Detectors For Neutrinoless Double Beta Decay And Other New Physics Searches, *Phys. Rev. D* **104**, 112007 **(2021)**

(SELECTED) RESEARCH EXPERIENCE

Graduate Research Experience (Departments of Physics) Case Western Reserve University

2020-2025
Cleveland OH, USA

- Designed and built gas-based proportional counters with analog–digital readout chains.
- Performed waveform analysis using Python and ROOT; improved detector calibration by ~20%.
- Presented at APS, SLAC workshops, and invited conferences.

Summer Intern Update International

2020-2025
Cleveland OH, USA

- Evaluated laser-photodiode combinations for precision measurement systems.

HONORS & AWARDS

- **Award for Outstanding Leadership** – Department of Physics and The School of Graduate Studies, CWRU, OH (2025)
- **Distinguished Student Travel Award** – American Physical Society Summit, CA (2025)
- **Poster Award: “Most Intriguing Title”** – Coordinating Panel for Advancing Detectors (CPAD) Workshop, TN (2024)
- **Travel Grant for Early-Career Researchers** – Oak Ridge National Laboratory & UT Knoxville, TN (2024)
- **Dignity of Difference: Student Leadership Award** – Division of Student Affairs, CWRU, OH (2023)
- **JN Tata Endowment Scholar** – Merit-Based Scholarship for Doctoral Study, Mumbai, India (2023)
- **Outstanding Student Organizations Award** – Treasurer, Graduate Council of Arts and Sciences, CWRU, OH (2022)
- **INSPIRE Top 1% Award** – Ministry of Science and Technology, Government of India, Delhi (2014-19)
- **National Top 1% Award** – National Graduate Physics Examination, India (2017)

CONFERENCES & PRESENTATIONS

- **Invited Speaker**, Summer Conference on High Energy Physics & Astrophysics – Liquid Xenon TPCs for Rare Event Searches, Virtual (2025)
- **Contributed Talk**, American Physical Society Global Physics Meeting– Noble Liquid Detector Symposium, CA (2025)
- **Participant**, Summer School – SLAC National Accelerator Lab, Stanford, CA (2025)
- **Finalist**, Poster Competition – Innovation Week, Case Western Reserve University, OH (2024)
- **Invited Attendee**, Conference for Graduate Women & Gender Minorities in Physics – Washington, DC (2024)
- **Poster Presenter**, Coordinating Panel for Advanced Detectors (CPAD) – UT Knoxville & Oak Ridge National Lab, TN (2024)
- **Speaker**, 3-Minute Thesis (3MT) Competition – Case Western Reserve University, OH (2024)
- **Participant**, Workshop on Xenon Detector $0\nu\beta\beta$ Searches – SLAC National Accelerator Laboratory, CA (2023)
- **Attendee**, Cleveland Biomedical Trainee Alliance Conference – CWRU, OH (2023)
- **Panelist**, Graduate Student Scholarship and Creative Endeavors Grant Review Committee – CWRU, OH (2022)
- **Participant**, Tri-Institute Summer School on Elementary Particles (Virtual) – SNOLAB, Sudbury, Canada (2021)
- **Attendee**, International Neutrino Summer School (Virtual) – CERN, Geneva, Switzerland (2021)
- **Participant**, XVth Rencontres du Vietnam, International Plasma Physics School – Quy Nhon, Vietnam (2019)

LANGUAGES

English – Fluent | Hindi – Native | Sanskrit – Basic
